Newton's laws of motion: history

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In 1686 Sir Isaac Newton published his three laws of motion – laws that form the basis of 'classical physics'. The movement of our daily lives is well described by these laws. When we play sport, the laws predict the outcome. Rocket scientists use the laws to predict the paths of their rockets and space craft. Many physics experiments in schools investigate them.

Although Newton had formulated the laws of motion possibly as early as 1666, it was only in 1686 that he published his single greatest work, the *Philosophiae Naturalis Principia Mathematica* ('Mathematical Principles of Natural Philosophy', simply known as the 'Principia').

In it, Newton revealed his laws of motion and the law of universal gravitation. The results were applied to orbiting bodies like planets and satellites, projectiles like bullets and golf balls, pendulums in clocks, and free-fall near the Earth.

The first law of motion is about inertia: a body wants to keep doing what it is doing. This means that a moving body wants to continue moving in a straight line, while a stationary body wants to remain stationary.

The second law says that the acceleration of an object depends on the force acting on it and its mass. ($a = F / m$)

The third law states that every action force has an equal but opposite reaction force.

Newton demonstrated that heavenly bodies mutually attract one another. Further generalisation led him to the law of universal gravitation: every object has a gravity force that attracts other objects. As the distance between the two objects increases, the strength of the gravity force between them decreases.

The Royal Society was going to publish Newton's book, but withdrew due to lack of funds. The astronomer Edmund Halley (of Halley's comet fame), who was wealthy and thought highly of Newton, eventually paid for its publication in London. The Principia is recognised by some as the greatest scientific book ever written.

1 Why is Isaac Newton often called Sir Isaac Newton?

2 Why might Newton have waited twenty years before publishing his laws?

3 When and where was the book published?

4 Who paid for the publishing costs?

5 What are Newton's laws about?

6 What is the keyword of Newton's first law?

7 According to that law, what does a moving object want to do?

8 What does a stationary object want to do?

9 If an object is moving in a straight line, what does it want to do?

10 If an object is falling towards earth, what does it want to do?

11 What is the formula of Newton's second law?

12 If the force is increased, what happens to the size of the acceleration?

13 If the mass is increased, what happens to the size of the acceleration?

14 What are the forces in Newton's third law?

15 If the action force pushes forward, in which direction does the reaction force push?

16 If the action force measures five newtons, how big is the reaction force?

17 If three action forces, measuring 3 N, 4 N and 5 N, are pushing upwards, how big is the reaction force and in which direction is it acting?

18 Folklore has it that Newton came up with the idea of gravity when an apple fell from a tree and hit him on his head.

a If the earth's gravity attracted the apple, did the apple's gravity also attract the earth?

b If the apple moved towards the earth as it fell, did the earth also fall towards the apple?

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